

SDS210 Wildfires Project: A focus on Australian Bushfires

Introduction

The chosen project topic for this assessment was the Wildfires Project, with a focus on Australia. As there is near real time satellite data and an API to request said data, I wanted to focus on accessing current “semi-live” data. The aim of the analysis was to replicate a fire monitoring and alert system that would be able to be used by government agencies to assist in rapidly responding to potential fires.

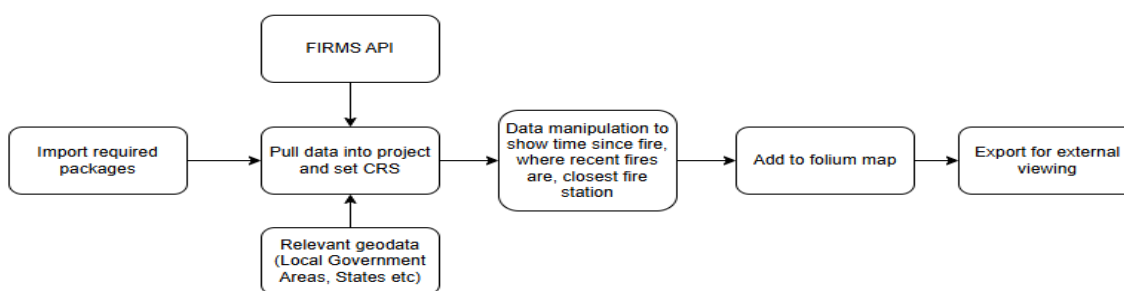
Specifically, I wanted to answer the following:

Where are the fire detections in Australia, and which fire service is best placed (i.e. closest) to deal with it?

Workflow and Implementation

The rough workflow that I started the project with developed into something more detailed and with further elements as the project progressed.

The initial workflow looked like this as I had it sketched out.



This then developed further by adding in the choropleth map and the identification of nearest Rural Fire Service (RFS) station.

On reflection, the project may have benefitted from loading all data in together at the start, so that CRS issues can be avoided and to make it easier for other users to replicate. As for the visualisation approach, it was an iterative process, with the maps being made and then I would perform numerous tweaks until I was happy with the output.

Results

The results of this project were three dynamic objects, two maps and a .html table. The aim was to identify where recent satellite-based fire detections were occurring, and [this interactive bushfire map](#) allows users to find locations and identify the confidence and time since detection. Due to four different sensors being included, we can also compare between sensors by toggling the layers on and off.

[This choropleth map](#) shows the number of fire detections per Local Government Area (LGA) in Australia, and [this .html table](#) shows a live update of fire detections within the last three hours and shows the distance to the closest RFS station.

Due to the dynamic nature of the input data, the observations should constantly be changing as satellites make new passes and the data is refreshed. Stagnant data means that the project isn't working properly.

Challenges and Reflection

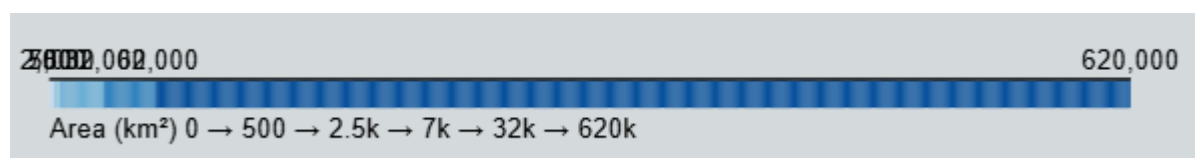
A project such as this requires the successful combination of a number of smaller programming elements and gave me a much better sense of how these smaller blocks that we learnt throughout the course can combine for much more complex tasks. There were of course challenges that arose during the process.

Function: “add_time_since_acq” (code block 5)

Creating this function was a challenge for me as I found it hard to wrap my head around exactly how the function was affecting the dataframes within. I also managed to jumble some of the calculations which led to incorrect results. It was therefore rewarding when I could apply it to each of the dataframes for clean results.

Folium Maps

The interactive outputs of the Folium maps are central to this project. And although Folium is an incredible package, they were however sometimes rather finicky to present in the ways I wanted to. For example, I still haven't been able to set a scale that is useful for the viewer, while also having the choropleth not be completely washed out. From my research I believe there are ways to edit and beautify this, but I ran out of time. See below for example.



Data Handling

Data proved to be often tricky to work with correctly. From the start, getting the API call to consistently work took time until I went and read the FIRMS guidance and used their code examples. I also was trying to use a 500MB .gpkg for my choropleth map which made everything very slow. To address this, I found the shapefile equivalent which was smaller, and then simplified it in the project to further reduce size.

Variable naming

Too much time was spent trying to get my choropleth map working as I had named two separate variables `modis_count` and `modis_counts` and was constantly getting them confused.

Repository Link & AI Statement:

https://github.com/liamedgeworth/Wildfires_SDS210.git

I used AI (Claude Sonnet 4.6) for assistance during the project. Predominantly usage was for debugging and providing information on how certain code blocks (particularly from the SDS210 lessons) functioned so I could understand how to replicate those code chunks for my project.

The element where I relied on it the most was for help with getting Git Actions to work, as I had no idea how this system functioned and the workflow documentation from Git did not seem to show the specific details of how to get it to work (e.g. where the workflow file should be located). The code and instructions provided by Claude were wholly used by me, however I understand now how the environment and workflow files work and re-run the notebook.

All final writing, logic, and project design was created by me.